

Question ID: ce508fb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The functions f and g are defined by the given equations.

$$f(x) = 3 + | - 2x - x^2 |$$

$$g(w) = | \frac{-w}{w-1} | - w + 5$$

If $f(-4) = c$, where c is a constant, what is the value of $g(c)$?

Correct Answer: -4.9, -49/10

Rationale

The correct answer is -4.9 . The value of $f(-4)$ is the value of $f(x)$ when $x = -4$. Substituting -4 for x in the equation $f(x) = 3 + | - 2x - x^2 |$ yields $f(-4) = 3 + | - 2(-4) - (-4)^2 |$, or $f(-4) = 3 + | - 8 |$, which is equivalent to $f(-4) = 3 + 8$, or $f(-4) = 11$. Since it's given that $f(-4) = c$, it follows that $c = 11$ and the value of $g(c)$ is the value of $g(11)$. Substituting 11 for w in the equation $g(w) = | \frac{-w}{w-1} | - w + 5$ yields $g(11) = | \frac{-11}{11-1} | - 11 + 5$, or $g(11) = | - 1.1 | - 6$, which is equivalent to $g(11) = 1.1 - 6$, or $g(11) = -4.9$. Note that -4.9 and -49/10 are examples of ways to enter a correct answer.